

SENSEI: A Simple, Elegant, and Nimble Subtitle Editing Interface

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1 MOTIVATION

Automatic subtitling keeps improving, but broadcast-ready output still needs human validation against reading speed, line length, lines-per-block, and text accuracy. Existing editors are either generic and free, or polished and closed. SENSEI closes that gap.

2 KEY CONTRIBUTIONS

- **Open by design.** Apache-2.0, self-hostable, web-based — no local install, usable from any browser.
- **Natively bilingual.** source and target are first-class, reactively synchronized tracks.
- **Compliance built in.** live, per-block, user-configurable monitoring of every formatting constraint.

3 SYSTEM ARCHITECTURE

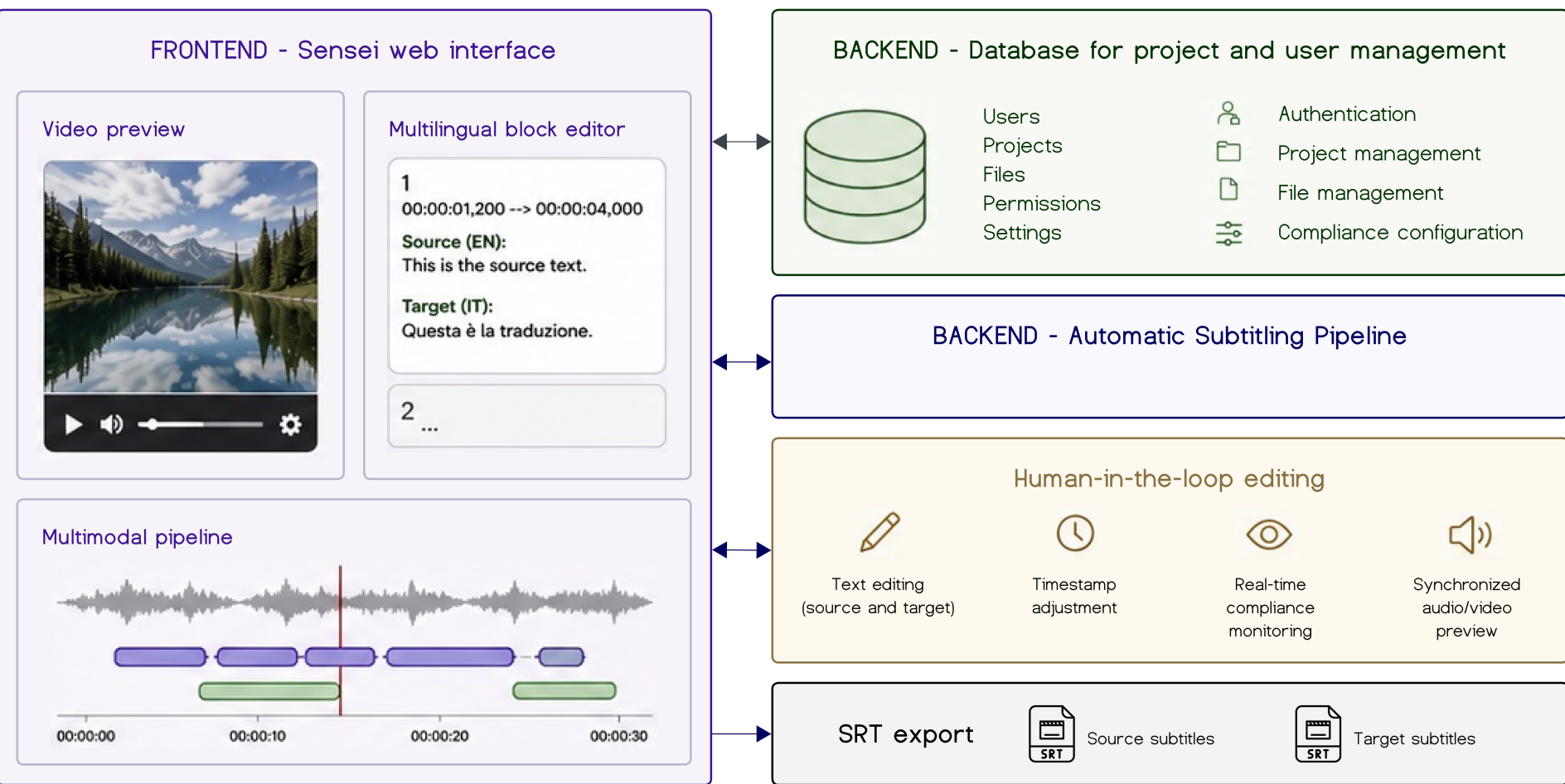
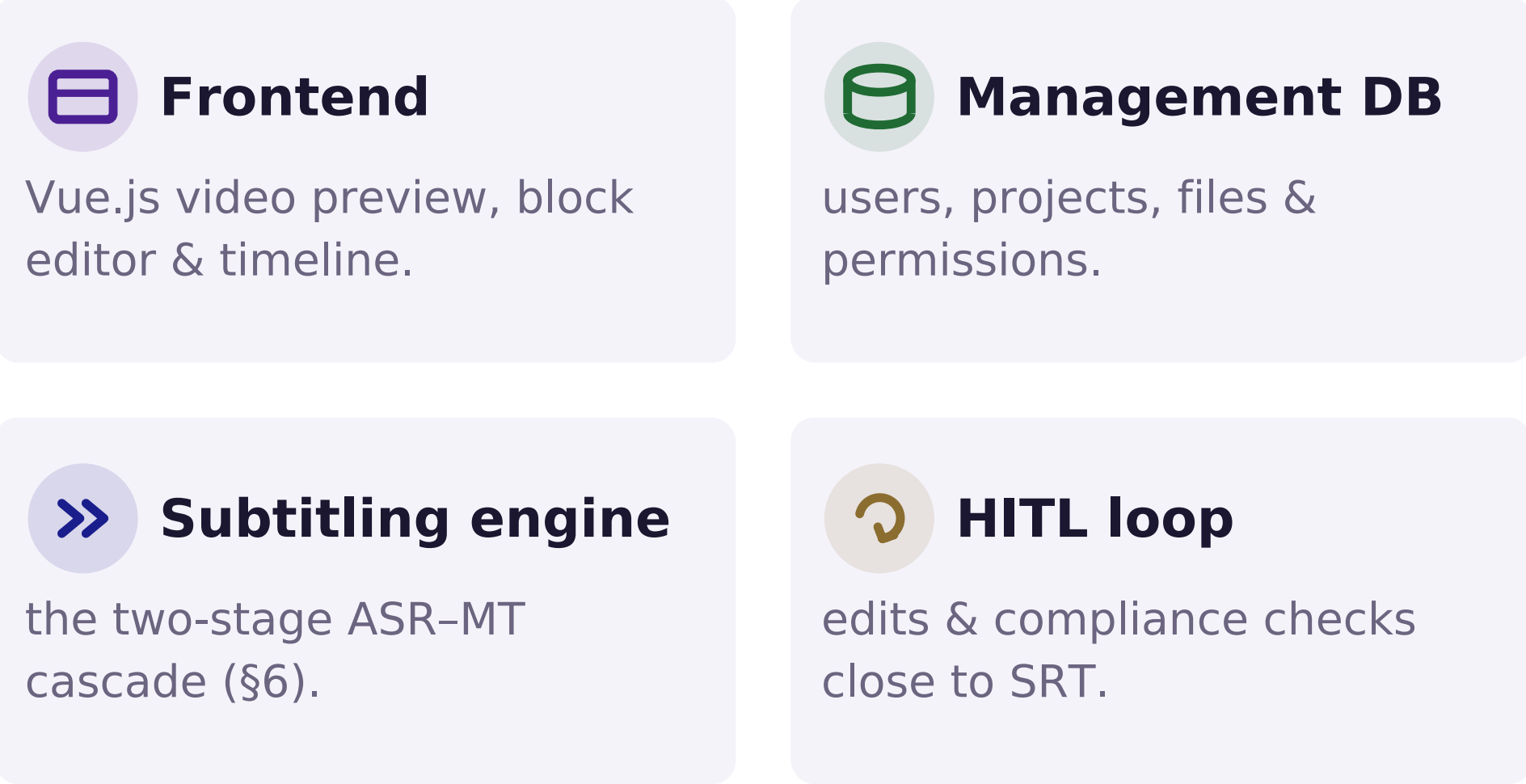


Figure 1. Overall architecture of the SENSEI ecosystem.



8 HOW DOES SENSEI COMPARE?

The only environment combining automatic generation, bilingual tracks and compliance-aware editing under a fully open license:

Functions	SENSEI	MateSub	HappyScribe	Aegisub	Subtitle Edit
Automatic subtitle generation	✓	✓	✓	□	✓
Merge	✓	✓	✓	✓	✓
Add	✓	✓	✓	✓	✓
Delete	✓	✓	✓	✓	✓
Duplicate	✓	□	□	✓	□
Edit text	✓	✓	✓	✓	✓
Edit timestamp (keyboard)	✓	✓	✓	✓	✓
Edit timestamp (mouse)	✓	✓	✓	□	✓
Fullscreen video	✓	□	✓	□	✓
Resizable interface	✓	open/close sidebar only	sidebar only	✓	✓
Multi-language	✓	□	separate environments	□	□
Load SRT file	✓	✓	✓	✓	✓
Playback of single subtitle	✓	✓	□	✓	vertical sidebar only

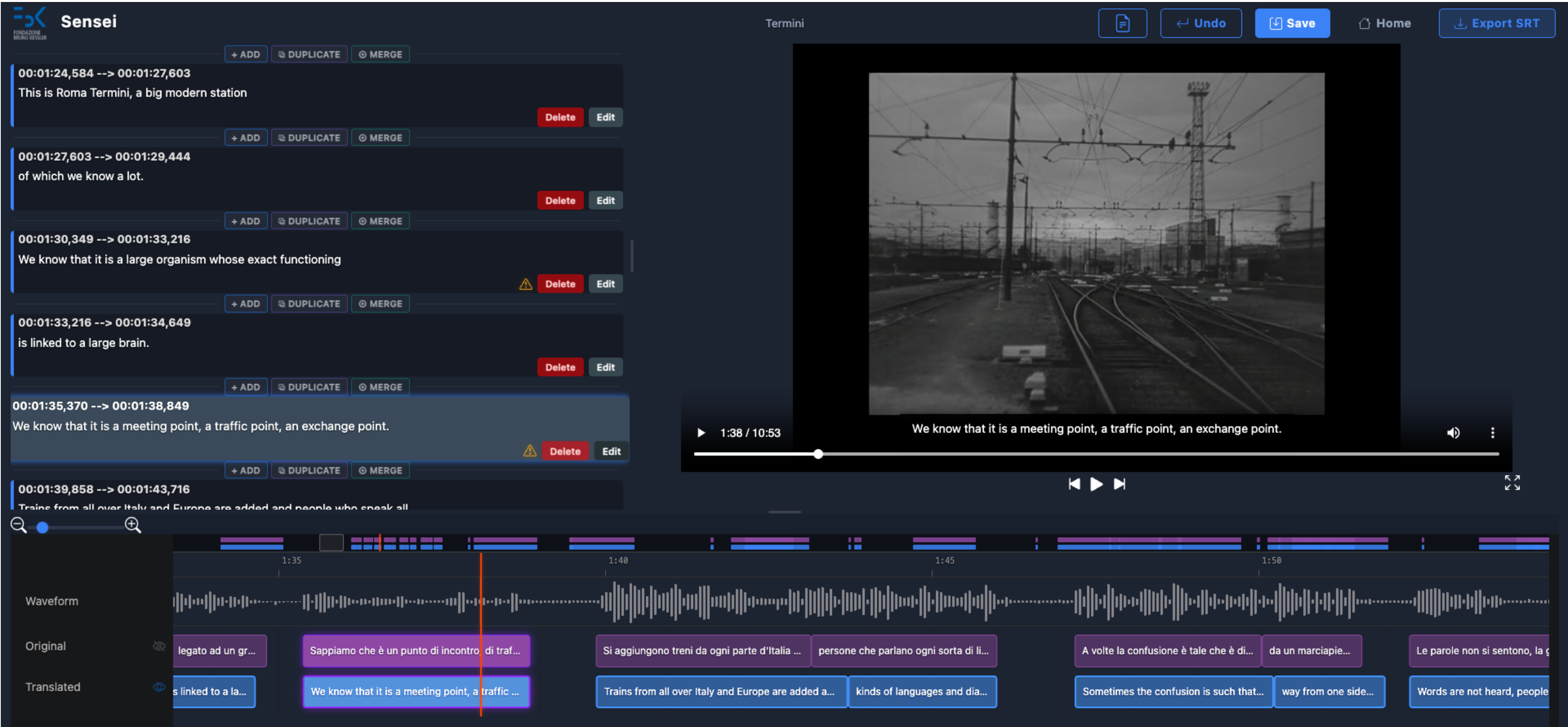
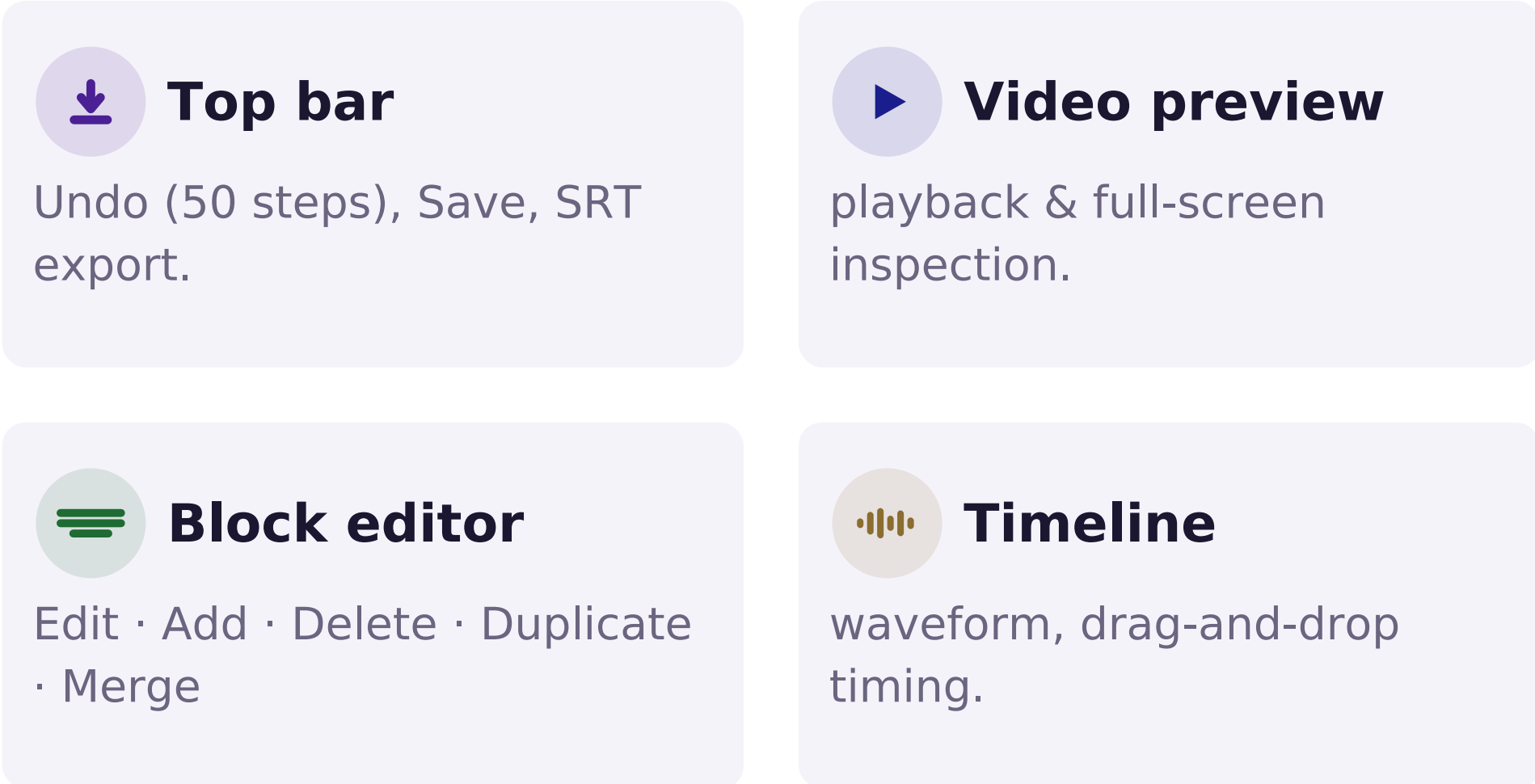
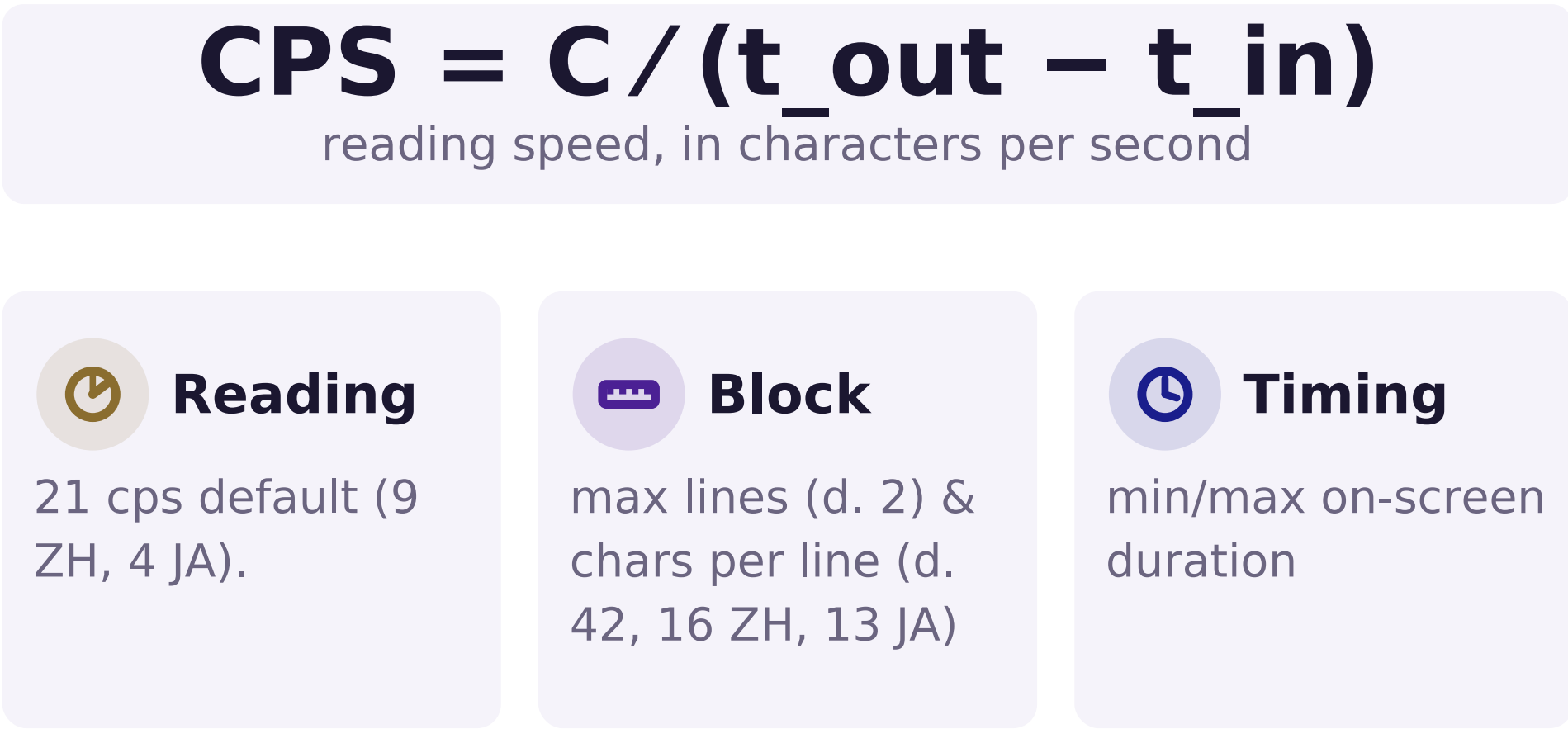


Figure 3. The SENSEI workspace, fully synchronized live.

4 THE EDITING INTERFACE



5 COMPLIANCE AT A GLANCE



All thresholds are user-editable; violations are flagged live.

6 SUBTITLING PIPELINE

A two-stage ASR-MT cascade, built entirely from freely available components.

STAGE 1 — BASELINE



STAGE 2 — SENTENCE-AWARE REFINEMENT



7 EVALUATION HIGHLIGHTS

- IWSLT 2026 Subtitling task, 3 domains, up to 5 languages: the sentence-aware cascade consistently beats the baseline on BLEU, ChrF and BLEURT.
- On ITV test2023: competitive with AppTek's commercial system, ahead of all previous FBK submissions (SubER).